

ZAHEER ABBAS

Senior AI/ML Engineer · Multi-Agent Systems Architect · LLM & Generative AI Specialist · Computer Vision Expert

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LinkedIn | Hugging Face | Remote • Contract • Hybrid

PROFESSIONAL SUMMARY

Accomplished AI/ML Engineer with **3+ years of hands-on production experience** architecting and deploying scalable, enterprise-grade intelligent systems that generate measurable business outcomes. Specialises in **LLM fine-tuning (LoRA/QLoRA/RLHF)**, **Retrieval-Augmented Generation (RAG)**, **multi-agent agentic orchestration**, real-time computer vision, and end-to-end MLOps. Proficient across the complete modern AI stack — PyTorch, LangChain, LlamaIndex, HuggingFace Transformers, vector databases (Pinecone, Chroma, Weaviate, FAISS), MLflow, Docker, and Kubernetes. Consistently delivers **89–92% model accuracy**, processes **60,000+ records and 5,000+ images monthly**, and reduces manual effort by up to **60%** through autonomous AI orchestration. Adept at converting cutting-edge research — RLHF, ReAct, chain-of-thought prompting, tool-use, and agentic planning — into robust, high-performance production systems.

CORE TECHNICAL SKILLS

LLMs & Gen AI:	GPT-4o · Claude 3 · LLaMA 3 · Mistral · Gemini · RLHF · LoRA · QLoRA · Prompt Engineering · Chain-of-Thought · Few-Shot · ReAct · Tool Use · Prompt Chaining · Instruction Tuning
Agentic & Multi-Agent:	LangChain · LlamaIndex · CrewAI · AutoGen · Agentic Workflows · Inter-Agent Communication · Task Decomposition · Memory & Planning · Autonomous Pipelines · Tool Orchestration
RAG & Vector Search:	RAG Pipelines · Semantic Embeddings · Hybrid Search · Pinecone · Chroma · Weaviate · FAISS · Document Intelligence · Knowledge Graphs · Contextual Retrieval
Deep Learning & ML:	PyTorch · TensorFlow · Keras · HuggingFace Transformers · XGBoost · scikit-learn · LSTM · Prophet · GANs · Distributed Training · Federated Learning · Model Quantization
Computer Vision:	YOLOv8 · OpenCV · MobileNetV2 · ResNet · TensorRT · Real-time Inference · Object Detection · Image Segmentation · Edge Deployment · Video Analytics
NLP & Arabic AI:	AraBERT · spaCy · BERT · Transformers · Semantic Similarity · Text Summarization · NER · Multilingual NLP · Clinical NLP · Islamic Text Analysis · Web Scraping
MLOps & Cloud:	MLflow · Docker · Kubernetes · CI/CD · GitHub Actions · FastAPI · Flask · Streamlit · AWS · Azure · GCP · Hugging Face Hub · Model Versioning · Feature Stores
Data Engineering:	Python · SQL · NoSQL · Firebase · Pandas · NumPy · Feature Engineering · EDA · Time-Series Forecasting · ETL Pipelines · Redis Caching · Web Scraping
Mobile & Full-Stack:	Flutter · Dart · JavaScript · REST APIs · Firebase Auth · Firestore · Android · Cross-platform AI Integration · Async State Management

PROFESSIONAL EXPERIENCE

Senior AI / Machine Learning Engineer

Jan 2023 – Present

Freelance — Independent Consultant | Remote

- Architected and deployed a **multi-domain AI platform** integrating LLM-powered agents, traffic prediction, energy forecasting, and real-time computer vision, processing **60,000+ structured records** and **5,000+ images monthly** via fully automated ETL pipelines at **99.5% uptime**.
- Designed end-to-end **production RAG systems** leveraging LangChain, Chroma, and FAISS with state-of-the-art embedding models, enabling semantic search and document intelligence at **sub-200ms query latency**.
- Fine-tuned open-source LLMs (LLaMA 3, Mistral) using **LoRA/QLoRA** on domain-specific corpora; incorporated RLHF feedback loops to achieve a **25% performance uplift** over baseline model benchmarks.
- Orchestrated a **40+ agent agentic ecosystem** (CrewAI, AutoGen) with tool-use, episodic memory, and inter-agent communication protocols, achieving a **60% reduction** in manual task-completion time through autonomous AI orchestration.
- Built and productionised a **YOLOv8 real-time vehicle detection system** attaining **87% mAP** on complex urban datasets with **sub-100ms inference latency**; optimised via TensorRT for edge deployment.
- Developed LSTM and Prophet time-series forecasting models, reducing prediction error by **23%** vs. baseline; engineered FastAPI inference microservices handling **100+ concurrent requests** at 99.5% availability.
- Integrated RLHF-style reward modelling and chain-of-thought prompting into LLM production pipelines, improving output quality and boosting user engagement by **35%**.

Freelance — Independent Consultant | Remote

- Integrated **on-device AI inference pipelines** into Flutter applications via optimised FastAPI backends, reducing data-synchronisation latency by **40%** through advanced async state management.
- Implemented secure Firebase Authentication, Firestore, and real-time database infrastructure for **3+ production applications** serving **500+ active users** at **99.9% availability**.
- Delivered cross-platform NLP chat assistants and computer-vision features using quantised PyTorch models and TFLite inference, improving AI-feature responsiveness by **50%**.

Full-Stack Developer — AI System Integrations

Sep 2023 – Present

Freelance — Independent Consultant | Remote

- Engineered scalable REST APIs using Flask and FastAPI capable of handling **10,000+ daily requests** with average response times under **200ms**; implemented Redis caching for **45% faster data retrieval**.
- Orchestrated seamless integration of production ML models and RAG pipelines into user-facing web applications, enabling real-time AI predictions for mission-critical business workflows.
- Established CI/CD pipelines with GitHub Actions and Docker, cutting deployment cycles by **30%** and ensuring reproducible model versioning across environments.

KEY PROJECTS**AI Hadith Authentication System****Lead Developer & Researcher | University FYP + Freelance | 2025 – Present**

- Conceived and engineered a **state-of-the-art Hadith authenticity classification system** capable of predicting whether a Hadith is *Sahih* (authentic), *Da'if* (weak), or *Mawdu'* (fabricated) — addressing a critical need in Islamic scholarship through the application of modern NLP.
- Fine-tuned **AraBERT** (Arabic BERT) on a purpose-built corpus of **34,000+ verified Hadith narrations**, achieving competitive classification accuracy across all three authenticity categories.
- Architected a robust **multilingual web-scraping pipeline** to collect, parse, and normalise Arabic Hadith data from authenticated Islamic APIs, digitised scholarly books, and peer-reviewed religious databases, ensuring training data integrity and provenance traceability.
- Designed an intelligent **chain-of-thought reasoning layer** that provides scholars and researchers with transparent, interpretable classification rationale — mapping predictions back to Isnad (chain of narration) attributes and semantic patterns in the Matn (text body).
- Deployed the system as a **FastAPI microservice with a Streamlit research dashboard**, enabling real-time Hadith verification queries with sub-300ms response time and supporting Arabic Unicode text input natively.

AI Urban Nexus — Smart City Intelligence Platform**Team Lead | End-to-End Development**

- Led the full-stack development of a comprehensive smart-city AI platform integrating LSTM, Prophet, XGBoost, and YOLOv8 for real-time traffic flow prediction, energy consumption forecasting, and public-safety analytics.
- Processed **60,000+ rows of structured data** and **5,000+ images** through automated ETL and validation pipelines; delivered a FastAPI + vector DB–powered analytics dashboard enabling **40% faster** data-driven decision-making for city planners.

ZMAR AI — Multi-Agent Agentic Orchestration Platform**Lead Developer | Autonomous AI Ecosystem**

- Designed and deployed a sophisticated multi-agent AI ecosystem with **40+ specialised agents** (LangChain, CrewAI) equipped with tool-use, episodic memory, and inter-agent communication, enabling collaborative problem-solving across diverse task domains.
- Achieved a **60% reduction in manual task-completion time** through autonomous orchestration; integrated RAG pipelines and semantic search for context-aware, long-horizon agent reasoning.

Arabic NLP Text Summarisation Engine**Solo Developer | Transformer-Based Abstractive & Extractive Pipeline**

- Developed a production-grade abstractive and extractive Arabic summarisation system using a fine-tuned **AraBERT** transformer; achieved **ROUGE scores competitive with global state-of-the-art benchmarks** on Arabic NLP leaderboards.
- Deployed a Hugging Face Inference–backed **FastAPI microservice** handling **1,000+ daily requests** with a **95% user satisfaction rate**.

Healthcare Predictive Analytics Platform**ML Engineer | Clinical Risk Assessment & Diagnostic Support**

- Built **HIPAA-compliant** patient risk-assessment and diagnostic-support models delivering **89–91% accuracy**, incorporating clinical NLP feature engineering and a federated-learning-ready architecture for data privacy.
- Reduced model inference latency by **50%** via TensorRT quantisation, optimised preprocessing pipelines, and asynchronous FastAPI serving.

ADVANCED LLM & PROMPT ENGINEERING EXPERTISE

- **4+ years** crafting production-grade prompts for GPT-4o, Claude 3, LLaMA 3, and Mistral using few-shot, chain-of-thought, ReAct, tool-use, and RAG techniques — consistently delivering **70%+ efficiency gains** in agentic workflow automation.
- Fine-tuned domain-specific LLMs with **LoRA/QLoRA** and RLHF reward modelling; achieved **25%+ task-performance improvement** over base models on custom evaluation benchmarks.
- Expert in **multi-agent system design** (LangChain, CrewAI, AutoGen), RAG architecture, vector database management (Pinecone, Chroma, FAISS), and large-scale semantic embedding pipelines.
- Delivered code-generation, content-automation, and document-intelligence agents at production scale with MLflow experiment tracking, Dockerised deployment, and Kubernetes horizontal scaling.

EDUCATION

Bachelor of Science in Computer Science

COMSATS University Islamabad | Expected Graduation: 2026

Relevant Coursework: Machine Learning · Deep Learning · Natural Language Processing · Neural Networks · Data Structures & Algorithms · Database Systems · Distributed Computing · Software Engineering

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- AI & Machine Learning Internship — **Certura Technologies** (Hands-on production ML project delivery)
- Digital Marketing & SEO Mastery — **DigiSkills Pakistan** (Data-driven growth strategies)
- Freelancing & Client Success — **DigiSkills Pakistan** (Professional project management)

ADDITIONAL INFORMATION

Languages: English (Fluent) · Urdu (Native) · Pashto (Native) · Punjabi (Conversational)

Availability: Full-Time · Remote · Contract · Hybrid